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NAZMUL HUDA BADHON

Deep Learning Researcher

Portfolio
LinkedIn
Google Scholar

Deep learning and machine learning researcher seeking opportunities to apply and grow expertise in AI, computer vision, and medical imaging.

EDUCATION

Bachelor of Science	Daffodil International University, Computer Science and Engineering CGPA: 3.88/4.00	Spring 2022–Fall 2025 Dhaka, Bangladesh
	Exchange Semester: Dongseo University, Computer Engineering GPA: 3.94/4.50 (93.25%)	Fall 2023 Busan, South Korea

RESEARCH INTERESTS

Deep Learning, Machine Learning, Medical AI, Computer Vision.

RESEARCH PUBLICATIONS

Research Articles

1. **Nazmul Huda Badhon**, Imrus Salehin, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, S M Noman, Nazmun Nessa Moon. TLCNN: tabular data-based lightweight convolutional neural network for electricity energy demand prediction. Global Energy Interconnection, Elsevier, 14 Nov 2025. ISSN: 2590-0358.
2. Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Imrus Salehin, Md. Sakibul Hassan Rifat, Faysal Ahmmed, Nazmun Nessa Moon. A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. MethodsX, 103680, Elsevier, 17 Oct 2025. ISSN: 2215-0161.

Review Articles

3. Imrus Salehin, Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Md. Sakibul Hassan Rifat, Nazrul Amin, Nazmun Nessa Moon. Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications. Engineering Reports 7.9: e70365, Wiley, 12 Sep 2025. ISSN: 2577-8196.

Data Articles

4. Imrus Salehin, Mahbubur Rahman Khan, Ummya Habiba, **Nazmul Huda Badhon**, Nazmun Nessa Moon. BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis. Data in Brief, 110083, Elsevier, 23 Jan 2024. ISSN: 2352-3409.

Book Chapter

5. Imrus Salehin, **Nazmul Huda Badhon**, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, Nazmun Nessa Moon. Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. Engineering Applications of AI for Demand Forecasting, Taylor & Francis. Accepted on 24 Apr 2025 (In press).

PROJECTS

1. Breast Cancer: Segmentation-Guided Hybrid Classification
   - Built a two-stage U-Net and CNN-based hybrid model on BUSI datasets.
   - Achieved strong performance on accuracy and ROC metrics.
2. Lung–Colon Histopathology: Image Enhancement & Transfer Learning
   - Applied preprocessing techniques on histopathology images and evaluated multiple transfer learning models.
   - Identified the best-performing model by accuracy and F1-score.
3. Medical Image Preprocessing & Quality Analysis with Vision Transformer
   - Designed preprocessing workflows for medical images and performed classification with ViT.
   - Achieved 87% testing accuracy and improved transformer performance.

TECHNICAL SKILLS

- **Programming**: Python, C, Java
- **Deep Learning**: Research-level implementation and model training
- **Frameworks**: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV
- **Data Analysis**: NumPy, Pandas, Matplotlib, Seaborn
- **Tools**: GitHub, Google Colab, LaTeX

AWARDS AND LEADERSHIP

Academic

- Global Korea Scholarship (GKS)–Fully funded exchange scholarship awarded by the Government of the Republic of Korea. Fall 2023
- Special Scholarship, DIUï Awarded for consistent academic excellence. Spring 2022ï Fall 2025

Research

- Research Award, Division of Research, Daffodil International University 2024
  Recognized for outstanding undergraduate research contribution.
- Peer Reviewer Recognition, Digital Health (SAGE Publishing). 2025

Leadership

- Team Leader, International Camp, Dongseo University, South Korea 2023
  Led a multicultural team of 20 students.
- Volunteer Internship, International Affairs, Daffodil International University 2024
  Assisted with student support and international programs.

REFERENCES

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